Reduced Optimal Control with PDE Constraint: Model, Discretization, and First-Order Methods

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Abstract

1 Problem Statement

We consider a linear-quadratic optimal control problem governed by the Poisson equation on a polygonal domain $\Omega \subset \mathbb{R}^2$ (in computations, typically $[0,1]^2$). A circular subregion $\omega \subset \Omega$ is tagged on the mesh and serves as the support of the control. The control acts as a volumetric source term restricted to ω:

$$-\Delta y = \chi_\omega \, u \quad \text{in } \Omega, \tag{1}$$

$$y = 0 \quad \text{on } \partial\Omega, \tag{2}$$

where χ_ω denotes the characteristic function of ω. We denote by $V = H_0^1(\Omega)$ the state space and by $U = L^2(\Omega)$ the control space. For a given target $y_d \in L^2(\Omega)$ and a regularization parameter $\beta > 0$, we minimize

$$J(y,u) = \tfrac{1}{2} \int_\Omega (y - y_d)^2 \, dx + \tfrac{\beta}{2} \int_\Omega u^2 \, dx. \tag{3}$$

The target is chosen as the pyramid-like profile

$$y_d(x) = 0.5 - \max\left(|x_1 - 0.5|,\, |x_2 - 0.5|\right), \tag{4}$$

which belongs to $L^2(\Omega)$. The control enters only on ω via χ_ω, so the effective source equals u on ω and vanishes on $\Omega \setminus \omega$. The PDE with homogeneous Dirichlet boundary conditions is well-posed for $u \in L^2(\Omega)$, yielding a unique weak solution $y \in V$ by Lax–Milgram.

2 Variational Formulation and Discretization

Weak formulation. Define the bilinear and linear forms

$$a(y,v) = \int_\Omega \nabla y \cdot \nabla v \, dx, \qquad b(u,v) = \int_\omega u \, v \, dx. \tag{5}$$

For any $u \in L^2(\Omega)$, the weak state equation reads: find $y \in V$ such that

$$a(y,v) = b(u,v) \quad \forall v \in V. \tag{6}$$

By coercivity of $a(\cdot,\cdot)$ on V and boundedness of $b(\cdot,\cdot)$, this admits a unique $y \in V$.

../plots/y_d_3d.png

Figure 1: 3D surface of the target profile y_d on Ω.

Finite elements. Let $\mathcal{T}_h$ be a conforming triangulation of Ω. We use the conforming P1 space $V_h \subset V$ with nodal basis $\{\varphi_i\}_{i=1}^n$. The control is represented in the same P1 basis but measured with the L^2 inner product ("control metric"). We assemble the matrices

$$A_{ij} = \int_\Omega \nabla\varphi_j \cdot \nabla\varphi_i \, dx, \quad M_{ij} = \int_\Omega \varphi_j \, \varphi_i \, dx, \quad B_{ij} = \int_\omega \varphi_j \, \varphi_i \, dx, \quad (M_U)_{ij} = \int_\Omega \varphi_j \, \varphi_i \, dx. \tag{7}$$

Here, A is the stiffness matrix for state/adjoint solves, M is the state mass matrix used to weight the tracking term, B maps control coefficients to the state RHS via restriction to ω, and M_U is the control-space mass matrix defining the L^2 inner product in U_h.

Boundary conditions. Homogeneous Dirichlet boundary conditions are imposed on the state/adjoint by modifying A and zeroing the RHS at boundary degrees of freedom when solving. Importantly, no boundary conditions are applied to B or M_U; these operators remain the unmodified L^2-based forms on Ω.

Discrete state equation and objective. With coefficient vectors $\mathbf{y}, \mathbf{u} \in \mathbb{R}^n$ and target coefficients $\mathbf{y}_d \in \mathbb{R}^n$, the discrete state equation and cost read

$$A\,\mathbf{y} = B\,\mathbf{u}, \tag{8}$$

$$J(\mathbf{y}, \mathbf{u}) = \tfrac{1}{2}(\mathbf{y} - \mathbf{y}_d)^\top M(\mathbf{y} - \mathbf{y}_d) + \tfrac{\beta}{2}\mathbf{u}^\top M_U\,\mathbf{u}. \tag{9}$$

../plots/mesh.png

Figure 2: Triangular mesh of Ω used for the P1 finite element discretization.

3 Reduced Formulation

State elimination. Eliminating the state via the discrete PDE constraint gives $\mathbf{y}(\mathbf{u}) = A^{-1}B\,\mathbf{u}$ and the reduced functional $F\colon \mathbb{R}^n \to \mathbb{R}$:

$$F(\mathbf{u}) = \tfrac{1}{2}(A^{-1}B\,\mathbf{u} - \mathbf{y}_d)^\top M(A^{-1}B\,\mathbf{u} - \mathbf{y}_d) + \tfrac{\beta}{2}\,\mathbf{u}^\top M_U\,\mathbf{u}. \tag{10}$$

Adjoint and gradient (continuous view). Introducing an adjoint variable $p \in V$ and the Lagrangian

$$\mathcal{L}(y,u,p) = \tfrac{1}{2}\|y - y_d\|^2_{L^2(\Omega)} + \tfrac{\beta}{2}\|u\|^2_{L^2(\Omega)} + a(y,p) - b(u,p), \tag{11}$$

the optimality system reads: state $a(y,v) = b(u,v)$, adjoint $a(v,p) = (y - y_d, v)$ for all $v \in V$, and variational gradient condition $(\beta u + \chi_\omega p, w)_{L^2(\Omega)} = 0$ for all $w \in U$. Hence, formally, $\beta u + p\,\chi_\omega = 0$ in Ω.

Adjoint and gradient (discrete view). In coefficients, the adjoint solves

$$A\,\mathbf{p} = M\,(\mathbf{y}(\mathbf{u}) - \mathbf{y}_d), \tag{12}$$

and the gradient with respect to the control space inner product induced by M_U is

$$\nabla F(\mathbf{u}) = M_U^{-1}\,B^\top \mathbf{p} + \beta\,\mathbf{u}. \tag{13}$$

The first-order optimality condition $\nabla F(\mathbf{u}_*) = 0$ is equivalent to

$$B^\top \mathbf{p}_* + \beta\,M_U\,\mathbf{u}_* = 0, \qquad A\,\mathbf{p}_* = M\,(\mathbf{y}(\mathbf{u}_*) - \mathbf{y}_d). \tag{14}$$

Hessian, convexity, and Lipschitz constant. The reduced Hessian applied to $\mathbf{v}$ is

$$H\,\mathbf{v} = M_U^{-1}\,B^\top A^{-1} M A^{-1} B\,\mathbf{v} + \beta\,\mathbf{v}. \tag{15}$$

Since A, M, and M_U are symmetric positive definite on their natural subspaces and B is linear, the quadratic form associated with H is strictly positive for $\beta > 0$; thus, F is strictly convex with a unique minimizer. In the U-metric, the gradient is Lipschitz with constant

$$L = \left\| M_U^{-1/2} B^\top A^{-1} M A^{-1} B\, M_U^{-1/2} \right\|_2 + \beta, \tag{16}$$

namely the spectral radius of the SPD term plus β.

4 Optimization Methods

We first rewrite the reduced problem as a quadratic optimization problem in the control coefficients. Using the discrete operators, the objective

$$F(\mathbf{u}) = \tfrac{1}{2}(A^{-1}B\,\mathbf{u} - \mathbf{y}_d)^\top M (A^{-1}B\,\mathbf{u} - \mathbf{y}_d) + \tfrac{\beta}{2}\,\mathbf{u}^\top M_U\,\mathbf{u}$$

expands to the unconstrained QP

$$\min_{\mathbf{u}\in\mathbb{R}^n}\ \tfrac{1}{2}\,\mathbf{u}^\top Q\,\mathbf{u} + \mathbf{c}^\top \mathbf{u} + \text{const}, \qquad Q = B^\top A^{-1} M A^{-1} B + \beta\, M_U, \quad \mathbf{c} = -\,B^\top A^{-1} M\,\mathbf{y}_d, \tag{17}$$

with constant term $\text{const} = \frac{1}{2}\,\mathbf{y}_d^\top M\,\mathbf{y}_d$. Since A, M, and M_U are SPD on their natural subspaces, Q is SPD, so the QP is strictly convex with unique minimizer characterized by the normal equations

$$Q\,\mathbf{u}_* = B^\top A^{-1} M\,\mathbf{y}_d. \tag{18}$$

Equivalence with reduced KKT. Rearranging the normal equations gives

$$\beta\, M_U\,\mathbf{u}_* \;+\; B^\top A^{-1} M\,(A^{-1}B\,\mathbf{u}_* - \mathbf{y}_d) \;=\; 0.$$

Defining the adjoint by $A\,\mathbf{p}_* = M\,(A^{-1}B\,\mathbf{u}_* - \mathbf{y}_d)$ (i.e., $\mathbf{p}_* = A^{-1}M\,(A^{-1}B\,\mathbf{u}_* - \mathbf{y}_d)$) yields the reduced optimality condition

$$B^\top \mathbf{p}_* + \beta\, M_U\,\mathbf{u}_* = 0,$$

which is exactly the KKT stationarity written in Section 3.

We solve the unconstrained, strictly convex reduced problem in the control space endowed with the L^2 inner product induced by M_U:

$$\min_{\mathbf{u}\in\mathbb{R}^n}\ F(\mathbf{u}) = \tfrac{1}{2}(A^{-1}B\,\mathbf{u} - \mathbf{y}_d)^\top M (A^{-1}B\,\mathbf{u} - \mathbf{y}_d) + \tfrac{\beta}{2}\,\mathbf{u}^\top M_U\,\mathbf{u}. \tag{19}$$

The control-space inner product and norm are

$$\langle \mathbf{a}, \mathbf{b}\rangle_U := \mathbf{a}^\top M_U\,\mathbf{b}, \qquad \|\mathbf{a}\|_U := \sqrt{\langle \mathbf{a}, \mathbf{a}\rangle_U}. \tag{20}$$

The gradient used by all methods is the Riesz-represented gradient in U:

$$\nabla F(\mathbf{u}) = M_U^{-1}\,B^\top \mathbf{p} + \beta\,\mathbf{u}, \quad \text{where } A\,\mathbf{p} = M\,(\mathbf{y}(\mathbf{u}) - \mathbf{y}_d),\ A\,\mathbf{y}(\mathbf{u}) = B\,\mathbf{u}. \tag{21}$$

Unless stated otherwise, we initialize at $\mathbf{u}_0 = \nabla F(\mathbf{0})$ (as in the code) and stop when $\|\nabla F(\mathbf{u}_k)\|_U \le$ `tol` or a maximum number of iterations is reached.

Algorithm 1: Gradient Descent in U with step $\alpha = 1/L$

```
Data: tolerance tol > 0, max_iter, model with ∇F, ‖·‖_U
Result: sequence (u_k) converging to a minimizer
u ← u_0;
;                                                    // default ∇F(0)
Estimate L by power iteration; α ← 1/L;
for k = 0, 1, ... do
    g ← ∇F(u); g_n ← ‖g‖_U;
    if g_n ≤ tol then
        break
    end
    u ← u − α g;
end
```

4.1 Gradient Descent with Fixed Step $\alpha = 1/L$

For an L-Lipschitz gradient in the U-metric, the fixed-step update

$$\mathbf{u}_{k+1} = \mathbf{u}_k - \alpha\, \nabla F(\mathbf{u}_k), \quad \alpha = 1/L, \tag{22}$$

produces a monotone decrease in F for convex F. In practice, L is estimated via a matrix-free power iteration on the Hessian action $H\,\mathbf{v} = M_U^{-1} B^\top A^{-1} M A^{-1} B\,\mathbf{v} + \beta\,\mathbf{v}$.

4.2 Barzilai–Borwein (Alternating BB1/BB2)

BB methods approximate inverse curvature along the last step by selecting the steplength from secant information in the U-inner product. Define $\mathbf{s}_k = \mathbf{u}_k - \mathbf{u}_{k-1}$, $\mathbf{d}_k = \nabla F(\mathbf{u}_k) - \nabla F(\mathbf{u}_{k-1})$, and the U-inner product $\langle \cdot, \cdot \rangle_U$. The two classical choices are

$$\alpha_k^{\mathrm{BB1}} = \frac{\langle \mathbf{d}_k, \mathbf{d}_k \rangle_U}{\langle \mathbf{s}_k, \mathbf{d}_k \rangle_U}, \qquad \alpha_k^{\mathrm{BB2}} = \frac{\langle \mathbf{s}_k, \mathbf{s}_k \rangle_U}{\langle \mathbf{s}_k, \mathbf{d}_k \rangle_U}. \tag{23}$$

We alternate BB1/BB2 per iteration and fall back to $\alpha_k = 1$ if $\left|\langle \mathbf{s}_k, \mathbf{d}_k \rangle_U\right|$ is numerically tiny. The update is $\mathbf{u}_{k+1} = \mathbf{u}_k - (1/\alpha_k)\,\nabla F(\mathbf{u}_k)$.

4.3 Nesterov Acceleration with Known (m, L) (Constant Parameters)

For the strongly convex quadratic QP with Hessian Q in the U-metric, let $L = \lambda_{\max}$ and $m = \lambda_{\min}$ denote the largest/smallest generalized eigenvalues of the pencil (Q, M_U), equivalently the spectrum of $H = M_U^{-1/2} Q M_U^{-1/2}$. Set

$$\alpha = 1/L, \qquad \kappa = L/m, \qquad \beta = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}.$$

Then the constant-parameter Nesterov scheme reads

$$\mathbf{y}_k = \mathbf{u}_k + \beta\,(\mathbf{u}_k - \mathbf{u}_{k-1}), \qquad \mathbf{u}_{k+1} = \mathbf{y}_k - \alpha\,\nabla F(\mathbf{y}_k), \tag{24}$$

and achieves a linear rate $\|\mathbf{u}_k - \mathbf{u}_*\|_U \le C\,\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k$. In practice, we compute (L, m) by solving the generalized eigenproblem (Q, M_U) with $Q\,v = \lambda\, M_U\, v$, where $Qv = B^T A^{-1} M A^{-1} B v + \beta M_U v$ is applied matrix-free via $Qv = M_U\, H v$.

Algorithm 2: Barzilai–Borwein in U (alternating BB1/BB2)

```
Data: tol > 0, max_iter, model with ∇F, ⟨·,·⟩_U, ‖·‖_U
Result: sequence (u_k)
u_{-1} ← 0; g_{-1} ← ∇F(0);
u_0 ← u_0;
g_0 ← ∇F(u_0);
;                                                    // default g_{-1}
for k = 0, 1, ... do
    if ‖g_k‖_U ≤ tol then
        break
    end
    s_k ← u_k − u_{k−1}; d_k ← g_k − g_{k−1}; sd ← ⟨s_k, d_k⟩_U;
    if |sd| < ε then
        α_k ← 1;
    else
        if k is even then
            α_k ← ⟨d_k, d_k⟩_U / sd;
        else
            α_k ← sd / max{⟨s_k, s_k⟩_U, ε};
        end
    end
    u_{k+1} ← u_k − (1/α_k) g_k;
    u_{k−1} ← u_k; g_{k−1} ← g_k;
    u_k ← u_{k+1}; g_k ← ∇F(u_k);
end
```

Algorithm 3: Nesterov with known (m, L) (constant β)

```
Data: tol > 0, max_iter, model with ∇F, M_U, Hessian action H
Result: sequence (u_k)
Estimate (L, m) from generalized eigenpairs of (Q, M_U); α ← 1/L; κ ← L/m;
 β ← (√κ − 1)/(√κ + 1)
u_{-1} ← u_0; u ← u_0;
for k = 0, 1, ... do
    y ← u + β(u − u_{k−1});
    g ← ∇F(y); g_n ← ‖g‖_U;
    if g_n ≤ tol then
        break
    end
    u_{k+1} ← y − α g;
end
```

4.4 Remarks on Metrics and Implementation

All inner products and norms above are in the control metric induced by M_U. This choice ensures that steplengths and convergence tests reflect the natural topology of the control space (consistent with regularization). The adjoint reuse of A facilitates efficient linear solves and preconditioning.

5 Numerical Results

We evaluate three first-order methods on the external-mesh reduced model with control region tag $\omega = 2$ and Tikhonov parameter $\beta = 10^{-3}$. The initialization is $\mathbf{u}_0 = \nabla F(\mathbf{0})$ and the stopping criterion is $\|\nabla F(\mathbf{u}_k)\|_U \leq \texttt{tol}$ or a cap on iterations.

The Lipschitz constant used by fixed-step methods is estimated via power iteration, yielding $L \approx 1.0010 \times 10^{-3}$. For constant-parameter Nesterov, the generalized eigenvalues of the Hessian in the U-metric satisfy $m \approx 1.0000 \times 10^{-3}$ and $L \approx 1.3072 \times 10^{-3}$.

Method	Final cost $F(\mathbf{u}_k)$	Iterations
BB	0.0162550	4
GD $(1/L)$	0.0162550	12
Nesterov (m, L)	0.0162550	6

Table 1: Final objective value and iteration counts. All methods reach the same optimum; BB attains it in the fewest iterations.

../plots/grad_norm.png

../plots/cost.png

Figure 3: Convergence of gradient norm (left) and objective value (right). The control-space norm $\|\cdot\|_U$ is induced by M_U.

Overall, all reported methods converge to the same minimizer of the strictly convex reduced problem, corroborating the quadratic structure. In our run, Barzilai–Borwein required only 4 iterations to reach tolerance, while fixed-step GD and constant-parameter Nesterov took 12 and 6 iterations, respectively.

../plots/suboptimality.png

Figure 4: Suboptimality $F(\mathbf{u}_k) - F(\mathbf{u}_*)$ on a logarithmic scale, highlighting linear convergence for strongly convex quadratics and relative speed differences among methods.

6 Mesh Independence

We assess the sensitivity of the optimizer and solution with respect to the mesh resolution. Triangular meshes of the unit square with a circular control region ω are generated via Gmsh for global sizes $h \in \{0.06, 0.04, 0.03, 0.02, 0.015, 0.012, 0.01\}$. On each mesh, we solve the reduced problem with $\beta = 10^{-3}$ and $\omega = 2$ (the physical tag of the circle) and compute the unique minimizer $u_*^{(h)}$ by solving the normal equations $Qu = B^\top A^{-1} M\, y_d$ using CG.

For each h, we record the final objective value $F(u_*^{(h)})$, the control-space norm $\|u_*^{(h)}\|_U$, the state L^2-norm $\|y(u_*^{(h)})\|_{L^2(\Omega)}$, and also the CG iteration count used to compute $u_*^{(h)}$ via the normal equations. Figure **??** shows that $F(u_*^{(h)})$ stabilizes as h decreases, indicating mesh independence of the discretized reduced problem and its optimizer. The accompanying norms remain consistent across refinements.

h	DOFs	$F(u_*^{(h)})$	$\|u_*^{(h)}\|_U$	$\|y(u_*^{(h)})\|_{L^2}$
height				

Table 2: Per-mesh summary of objective and norms. DOFs denotes the number of P1 degrees of freedom.

../plots/distance_to_opt.png

Figure 5: Distance to the optimizer in the control metric, $\|\mathbf{u}_k - \mathbf{u}_*\|_U$, confirming convergence of all methods to the unique minimizer and illustrating rate differences.

h	CG iters (u_*)	BB iters	GD($1/L$) iters	Nesterov(m, L) iters
height				

Table 3: Iterations per mesh: CG iterations to obtain u_* by normal equations, and iteration counts for first-order methods.

../plots/u_opt.png

../plots/y_opt.png

Figure 6: Optimized control u_* (left) supported on ω and the corresponding state $y(u_*)$ (right).

../plots/mesh_independence_cost_vs_h.png

../plots/mesh_independence_norms_vs_h.png

Figure 7: Mesh independence study: objective value $F(u_*^{(h)})$ versus mesh size h (left), and norms $\|u_*^{(h)}\|_U$ and $\|y(u_*^{(h)})\|_{L^2}$ versus h (right).